tasks

Tasks: Authored AI/IT Knowledge Areas with Lessons, Tests and Video Anchors

Input: Design documents from `specs/004-authored-ai-areas/`
Prerequisites: plan.md, spec.md, research.md, data-model.md, contracts/, quickstart.md

How to read this file

Tasks marked `[x]` are DONE and were verified with a captured run; the evidence is on the task line. They are kept rather than deleted because this feature was built by four parallel streams during the session that specified it, and a task list that hid the completed work would misrepresent both what remains and how the remainder must be verified.

A checked box here means a command was run and its exit code observed. It does not mean the capability is live — several completed tasks are verified in-process and remain unverified over HTTP. Those are checked here and listed again under Phase 8, which exists precisely so an in-process pass is never mistaken for a served one.

Task Format

`[ID] [markers] [Story] Description with file path`

Markers: `[P]` parallelizable · `[TDD]` RED-GREEN-REFACTOR · `[REVIEW]` review before consumers are built · `[SUBAGENT]` delegable

Story labels: `[US1]`..`[US6]` map to the user stories in spec.md.

Path Conventions

Paths are repository-relative from the umbrella root. The two roots that matter: `workshop/` (the private module) and `submodules/curriculum-kit/` (the reusable library, project-unaware).

Phase 1: Setup (Shared Infrastructure)

Purpose: The reusable shape and the strict content format exist before anything is authored against them.

- [x] T001 [REVIEW] Create the reusable curriculum library in `submodules/curriculum-kit/` — `Catalog → Area → Lesson → Material` plus `Area.Assessment`, with `VideoAnchor{chapterId,startMillis,endMillis,transcriptAnchor}` (FR-006, FR-013, FR-014). **Evidence**: 82 tracked files (source); `go test ./...` ok (in_process).
- [x] T002 Keep the library project-unaware by construction in `submodules/curriculum-kit/go.mod` — an empty require set, so a project import would not compile (FR-?). **Evidence**: compiler-enforced, not convention.
- [x] T003 [P] Implement `DecodeDocument` with `DisallowUnknownFields()` in `submodules/curriculum-kit/pkg/curriculum/document.go` (FR-?). **Evidence**: a misspelled key fails the load loudly instead of dropping its material silently.
- [x] T004 [P] Make `Area.Assessment` a pointer so “under construction” is representable, and have `Validate` emit `CK021` for it rather than letting the absence pass unnoticed (FR-008, SC-004a). **Evidence** (population: unstated): 37 CK021 findings raised, none suppressed.

Execution notes: T003 is the load-bearing decision of this phase. Everything authored later either decodes or fails at load; there is no partial-success path.

Phase 2: Foundational (Blocking Prerequisites)

Purpose: The single publication decision, and the evidence base the authored set is selected from. **No user story work can begin until this phase is complete**, because every story's visibility derives from T007.

- [x] T005 Measure module evidence across **all** recordings in `workshop/pipeline/extract/measure_module_evidence.py` (FR-005, FR-005a, FR-005b, SC-016). **Evidence** (source): rc 0; 0 of 37 modules with zero evidence, 0 of 137 terms matching zero passages; 16 strong / 9 moderate / 12 thin.
- [x] T006 Enforce the already-written "title non-empty when published" rule in `workshop/pipeline/extract/publication_policy.py` **and** in the pipeline reconcile stage, so a re-run reproduces it (FR-003, FR-018, FR-018b, SC-001). **Evidence** (source): published 819 → 42; nothing deleted, withheld rows to `workshop/curriculum/unpublished-areas.jsonl`.
- [x] T007 [TDD] [REVIEW] Compute the publication decision **once** in `workshop/platform/backend/internal/api/area_publication.go` (`AreaPublicationOf`) and consume it from all three surfaces (FR-018, FR-018a, SC-001). **Evidence** (in_process): `verify-area-publication-consistency.sh` rc 0 (6 passed); `prove-area-publication-consistency.sh` rc 0 (7 passed / 7 mutations); Go `TestAreaListAndDetailAgree` asserts the **pair**, which is what was missing while both handlers' own tests passed.
- [x] T008 [P] Promote the 37 authored subjects through the real minting bridge in `workshop/pipeline/extract/promote_curriculum_areas.py`, supplying a lexicon evidence finder instead of the title-keyword default (FR-001, FR-002, FR-005a, SC-002). **Evidence** (source): rc 0, 37 published / 0 failed; reviews 2 → 39.
- [x] T009 [P] Keep `proposer` and `reviewer` distinct in the review store, both defaulting to the pipeline string (FR-002). **Evidence**: honest description of a run with no independent audit; `--reviewer` is passed only when a human has read them.

Checkpoint: The catalogue is honest about itself. **T007 is the checkpoint** — `areas[]` is now exactly the set whose detail answers 200, asserted end to end.

Phase 3: User Story 1 — A learner finds a subject they recognise (P1) MVP

Goal: Every catalogue entry is nameable by a person and every entry opens. **Independent Test**: Open the catalogue, read every visible entry, open each one.

- [x] T010 [US1] Serve `areas[]` as exactly the servable set, with `held_back[]`, the reason table and `total_areas` in the same response, in `workshop/platform/backend/internal/api/` (FR-018, FR-018b, SC-001). **Evidence** (population: unstated): 817 unopenable → 3.
- [x] T011 [US1] Disclose a derived title/tags/summary on a **refusal** where one is derivable, in the same handler (FR-018b). **Evidence**: the reported symptom was a page reading only "withheld pending review"; an area held back for a missing review has a perfectly good name.
- [x] T012 [P] [US1] Add `?include=held_back` so withholding never makes an area unreachable (FR-018b).
- [x] T013 [P] [US1] Emit `tags` as an empty array, never `null`, in the area wire type (FR-003, FR-019). **Evidence**: a null breaks the field the interface filters on.
- [x] T014 [US1] Derive titles, tags and summaries in `workshop/platform/backend/pkg/knowledge/presentation.go` (FR-003, FR-018b).
- [x] T015 [US1] Add a `tags` field to the client area model in `workshop/platform/frontend/src/app/core/knowledge.ts` — it models `terms` and **has no field for tags at all**, so tags are discarded before any component can ask for them (FR-019, SC-005). — EVIDENCE (source): `workshop/platform/frontend/src/app/features/areas/curriculum-model.ts:487,505` (`AreaCard.tags`, `normaliseAreaCard`) + `area-detail.component.ts:451` — PATH DRIFT: delivered in a second normaliser, not in `core/knowledge.ts`, which still drops `tags`; the

capability (FR-019) holds. Frontend unit suite 298/298 SUCCESS (in_process) 2026-09-08

- [x] T016 [US1] Render title, summary and tags on the area detail page in `workshop/platform/frontend/src/app/features/` — 3 of 4 render branches emit the raw identifier as the heading (SC-003, FR-003, FR-019, SC-005). — EVIDENCE (source): `workshop/platform/frontend/src/app/features/areas/area-detail.component.ts:111` single `<h1 data-testid="area-title">{{ displayTitle() }}</h1>`, `:442` title chain with the slug as LAST resort, `:446` summary, `:115-121,451` tags list. Frontend unit suite 298/298 SUCCESS (in_process) 2026-09-08

Checkpoint: A learner can open every listed area and read what it is.

Phase 4: User Story 2 — A learner works through an area and is tested (P1)

Goal: Lessons in order, a gated end-of-area test, a scored result. **Independent Test**: Read one area's lessons, take its test answering some right and some wrong, check the score and per-question feedback match.

Tests for User Story 2

> Written FIRST and verified to fail. The load-bearing one is T017.

- [x] T017 [TDD] [US2] Prove the assessment gate with **17 data mutations** in `workshop/platform/gates/prove-assessment-gate.sh`, including B4/M7 "the gate opens as soon as *any* lesson completes" — which passes B1, B2, B3, B5 and B7 and is caught **only** by B4 (FR-010, FR-012a, FR-028, SC-009). **Evidence** (in_process): rc 0, 17 passed. Requires the fixture's required set to be a **proper subset** of its lessons.
- [x] T018 [TDD] [US2] Add a leak check that fails if `correct_choices` or `explanation` appears on the taking path, in `workshop/platform/backend/internal/api/lessons_test.go` (FR-010, SC-009).
- [x]

T019 [P] [US2] Add vacuity refusals **B9/B10** — no lesson outside the required set, or a test gated on nothing, must exit **2**, never 0 (FR-027, FR-029).

Implementation for User Story 2

- [x] T020 [US2] Implement the six learning routes in `workshop/platform/backend/internal/api/lessons.go`, all behind the T007 decision, all keyed on the opaque `X-Session` header only (FR-006, FR-007, FR-009, FR-010, FR-012a, FR-013).
- [x] T021 [US2] Build the taking projection as a map with **no answer field at all** in the same file — not the domain type with fields omitted, so no future upstream field can leak through it (FR-010, FR-012a, SC-009).
- [x] T022 [P] [US2] Validate `?kind=` against the closed vocabulary and return **400** for an unknown kind (FR-013). **Evidence**: an empty list reads as “there are none”, which is a different and false statement.
- [x] T023 [US2] Implement the session progress store in `workshop/platform/backend/pkg/learning/` — one `Mutate` under a single lock, atomic temp-file+rename, a file separate from the reading-position store (FR-010, FR-012a). **Evidence**: a Get+Put cycle silently drops a concurrent completion.
- [x] T024 [US2] Build the authored learning catalog in `workshop/pipeline/extract/build_learning_catalog.py` → `workshop/curriculum/learning/` (FR-006, FR-007, SC-004). **Evidence** (source): 42 files, 339 lessons, 1,068 minutes; ids derived from the area ULID so catalog-wide collisions are structurally impossible.
- [x] T025 [US2] Resolve each bank’s `correct_index` to a choice **id** during the build (FR-012). **Evidence**: the kit rejects the index form by design.
- [x] T026 [US2] Exclude the 10 `flashcard` rows rather than mapping them onto graded items, and report the exclusion (FR-008). **Evidenc e**: mapping a study aid onto a test question is a silent promotion.
- [x] T027 [US2] Pass `-learning-catalog /opt/workshop/curriculum/learning` in `workshop/platform/compose.yml` (SC-006). **Evidence**

(source): the flag defaults to EMPTY and empty is a *determined* state — every learning route would answer `no_learning_catalog` cleanly after a restart, with nothing in any log to notice. YAML re-parsed; the path is inside the existing read-only bind, so no volume change.

- [ ] T028 [US2] Render the lesson list and lesson body with prev/next in `workshop/platform/frontend/src/app/features/`, using the same sorted slice the list uses (FR-007, SC-006).

- [x] T029 [US2] Render the test — availability message before completion, questions after, result after submission — in the same feature directory, showing `pass_percent` **before** the attempt begins (FR-008b, FR-009, FR-010, SC-017). — EVIDENCE (source): `workshop/platform/frontend/src/app/features/areas/area-test.component.ts:75-112` availability + `passPercent` shown before the attempt, `:169-181` result panel with the determinate/lower-bound split. Frontend unit suite 298/298 SUCCESS (in_process) 2026-09-08

Checkpoint: The full learner journey works against authored content.

Phase 5: User Story 3 — Jump to the exact moment in the recording (P2)

Goal: Follow a lesson reference; land on the recording at that moment with the transcript scrolled and the passage marked. **Independent Test**: Follow every reference in one area and confirm all four consumer obligations.

- [x] T030 [US3] Write the URL grammar down as the producer's authority in `workshop/platform/frontend/docs/time-links.md` (FR-014, FR-015).

- [x] T031 [US3] Build spans from engine-measured start/end times joined on passage id, in `workshop/pipeline/extract/build_area_timelinks.py` → `workshop/curriculum/area-timelinks.json` (FR-014, FR-016). **Evidence** (source): rc 0, 205 spans; the passage registry carries **only** an end time, so a start derived from it alone would be invented.

- [x]

T032 [US3] Emit `end` **only when `> t`** and `#p-` **only for a Crockford-shaped ULID**, in the material renderer (FR-014, FR-015). **Evidence**: the consumer ignores a malformed span, so producing one yields a link that renders correctly and does nothing.

- [x] T033 [US3] Emit `href: null` + `unresolved_reason` + intact anchor + a count for an unservable chapter (FR-016, FR-017, SC-007). **Evidence** (source): 205 of 205 resolved, 0 unresolved.

- [x] T034 [US3] Declare the chapter registry **from disk** (`workshop/chapters/`) rather than from the anchors themselves (FR-016, FR-029, SC-007). **Evidence** (source): deriving it from the anchors would make the resolution check vacuously pass; `ScanChapterDirs` returns exactly `01`, `02`, `02.01`.

- [x] T035 [US3] Implement the client-side time-link contract in `workshop/platform/frontend/src/app/core/timelink.ts` (FR-015).

- [x] T036 [US3] Seek the player to `t` and scroll the transcript to the `#p-` passage **without further scrolling**, marking it visually, in the transcript component (FR-015, SC-008). — EVIDENCE (source): `workshop/platform/frontend/src/app/features/transcript/transcript.component.ts:593` `#p-` parse, `:614-634` seek with `link.end`, `:691-693` `scrollIntoView`, `:631` `machine.seeked()` suspends follow so nothing scrolls further; `seek.spec.ts` + `follow-mode.spec.ts` green in the 298/298 run (in_process) 2026-09-08

- [x] T037 [P] [US3] Make the extent of a range discernible when `end` is present, not only its start (US3 scenario 3; FR-014). — EVIDENCE (source): `workshop/platform/frontend/src/app/core/playback.ts:88-95` `stopAt` bounds playback to `end` (and only when `end > t`); `features/chapters/recording-player.component.ts:84-87` `data-testid="span-end"` names the excerpt end. Frontend unit suite 298/298 SUCCESS (in_process) 2026-09-08

- [ ] T038 [P] [US3] Report an unresolvable target as unavailable rather than routing to a page that reports nothing found (FR-017).

Phase 6: User Story 4 — Materials, not only text (P2)

Goal: Illustrations, diagrams, schemes, graphs and in-place video segments. **Independent Test**: Open every lesson in one area; each declared material renders, is captioned, and is reachable without a mouse.

- [x] T039 [US4] Specify one diagram per area — type, the single idea, concrete nodes and edges — in `workshop/docs/training/diagrams/SPEC.md`, with deliberately smaller diagrams for the thin modules and the reason written down (FR-013).
- [x] T040 [US4] Build 8 SVGs in the house style extracted from `design-system/diagrams/`, **adding** `<title>`, `<desc>`, `role="img"` and `aria-label` — absent from 0 of 33 references (FR-013). **Evidence** (source): all 8 parse as XML, rc 0; no external font, image or script.
- [x] T041 [P] [US4] Serve `video` materials with `chapter_id`, `start_millis`, `end_millis`, `length_millis`, `transcript_anchor`, `chapter_slug` and `href` (FR-013, FR-014).
- [x] T042 [US4] Render materials in place with their captions in the lesson component (FR-013). — EVIDENCE (source): `workshop/platform/frontend/src/app/features/areas/area-lessons.component.ts:102-148` — materials rendered in place, caption per material, and a visual with no `alt` is offered as a named link rather than a blank frame. Frontend unit suite 298/298 SUCCESS (in_process) 2026-09-08
- [ ] T043 [P] [US4] Play a video segment in place, bounded to its stated range (FR-013).
- [ ] T044 [P] [SUBAGENT] [US4] Rasterise the 8 SVGs and check text fit at the served font metric — well-formedness and theming are verified; **text fit is not** (plan.md gap 7; FR-013).

Phase 7: User Story 5 — Vivid, legible, consistent (P2)

Goal: A varied palette that remains legible in both presentations. **Independent Test**: Sample the colours actually painted on each principal surface; measure variety across them and contrast within each pairing.

- [x] T045 [US5] Regenerate the served stylesheets from the widened token source in `workshop/platform/frontend/src/styles/`, then rebuild. **This is defect D6**: nothing had regenerated them, so a token-level widening never reached a visitor (FR-021, FR-023, SC-010). — EVIDENCE (served): `bash workshop/platform/gates/verify-served-palette.sh` rc **0** against the LIVE served bundle (`sample: live, http://127.0.0.1:8087`, 2 stylesheets, 208 surfaces graded): 7 distinct hue families, floor 6 — the widening reaches a visitor, so D6 is closed at the served layer
- [x] T046 [TDD] [SUBAGENT] [US5] Write a hue-family check that reads the **served** stylesheet — not the token source — asserting ≥6 distinct hue families across principal surfaces (SC-010). A check that reads the token source commits D6 inside the instrument meant to catch it (SC-010, FR-021, FR-023, FR-026). — EVIDENCE (served): `bash workshop/platform/gates/verify-served-palette.sh` rc **0** — reads the SERVED stylesheets over http://127.0.0.1:8087, not the token source; `PASS: 7 distinct hue family/families across principal surfaces (floor 6)`
- [x] T047 [TDD] [P] [SUBAGENT] [US5] Write a contrast check asserting the normal-text and non-text floors in **both** light and dark presentation, over the served bundle (SC-011, FR-022, FR-023, FR-026). — EVIDENCE (served): `bash workshop/platform/gates/verify-served-contrast.sh` rc **0** — `PASS: all 200 pairing(s) clear their floor, in each scheme independently` (8 unusable reported, not counted as passes)
- [x] T048 [P] [SUBAGENT] [US5] Pair T046 and T047 with data-driven mutation proofs, each carrying a control and a vacuity refusal that exits 2 (FR-027, FR-028, FR-029, SC-013, SC-014). — EVIDENCE (in_process): `bash workshop/platform/gates/prove-served-palette.sh` rc **0**, 8 passed / 0 failed / 8 mutations (M2 and M3 are rc-2 vacuity refusals, N1 the non-vacuity control); `prove-served-`

`contrast.sh` rc **0**, 6 passed / 0 failed / 6 mutations (M0 control, M2 rc-2 vacuity refusal, M3 the light-passes/dark-fails split). Gate source byte-identical across each battery

- [ ] T049 [US5] Emit `corpus_revision` on the served catalogue and cite it in every interface claim (FR-023a, SC-018). **Analysis finding F7 — MISPLACED**: this is a catalogue concern owned by the backend stream, not a presentation one. Execute it with Phase 3 (US1); it is listed here only because the superspec blueprint's Track A carries a *bundle* build id, which is a different artefact, and this task must not be assumed covered by it.

Execution note: rebuild immediately before measuring and **state which build was measured**. A stream that measures another stream's output mid-write reads a transiently broken file — this happened during construction and the partition was not what failed; the timing was.

Phase 8: User Story 6 — Each module runs on its own (P3)

Goal: Clone either module alone, bootstrap it, reach a running system. **Independent Test**: Clone into an empty directory outside the umbrella on a host that has never held it.

- [x] T050 [US6] Add `workshop/scripts/bootstrap-standalone.sh` and `workshop/helix-deps.yaml` (FR-024, SC-012).
- [x] T051 [P] [US6] Add `verify-standalone-clone.sh` to both `workshop/scripts/` and `ai_interviewing/scripts/` (FR-024, FR-025, FR-026, SC-012).
- [x] T052 [US6] Record the `curriculum-kit` replace-target gap verbatim in `workshop/platform/backend/go.mod` (FR-024). **Evidence** (source): gaps 2 → 3, correctly — the gate working, not drift.
- [x] T053 [US6] **BLOCKED — operator decision.** Publish `submodules/curriculum-kit` as a git repository and mount it as a gitlink, then declare it in `helix-deps.yaml`. It is currently a plain directory with **no commit to pin**, so a truthful manifest entry cannot be written and a fabricated `ref:` would be a bluff (FR-024, SC-012). —

EVIDENCE (source): `git ls-files -s submodules/curriculum-kit` -> `160000 52f08af5abb396bce7d9679cb82eb0b9dd0246fc`; `.gitmodules:40-42` declares the path and url; `helix-deps.yaml:313-316` records `ref: 52f08af5abb3...`; `bash scripts/verify-manifest-pins.sh` rc **0** at 13 MATCH / 0 DRIFT / 0 UNDETERMINED. The blocker (no commit to pin) is gone

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Phase 9: Polish & Cross-Cutting — turning in-process passes into served ones

Purpose: Every task above marked `[x]` that was verified **in-process** is still a **2** over HTTP. This phase is where that changes, and it exists as its own phase so the distinction is never quietly dropped.

- [x] T054 Stop every writing stream, then restart once with `bash workshop/scripts/restart.sh` (a full down/up, never `compose restart`). **Never pass `--destroy-volumes` to `stop.sh`** — the `workshop-index` volume holds hours of transcription output (FR-023, FR-023a). — EVIDENCE (served): container `workshop-curriculum_platform_1` started 2026-09-08 08:01:16, `Up 4 hours (healthy)` at measurement, and it serves the six learning routes that exist only in the new binary (all 200/403/400 below) — a full down/up, not a `compose restart`. Volume `workshop-curriculum_workshop-index` intact (487880 crossref edges over 24394 sources at generation 6)
- [x] T055 Re-run `bash workshop/platform/gates/verify-server-unity.sh` against the live container. **Currently RED at `PASS=37 FAIL=6`** — all six new routes answering a plain-text 404 because the container runs the old binary. Against a server built from this tree the same gate is `PASS=43 FAIL=0 UNDET=0 DEBT=8` (FR-023, FR-026). — EVIDENCE (served): `bash workshop/platform/gates/verify-server-unity.sh` rc **0** — `PASS=43 FAIL=0 UNDET=0 DEBT=8`, exactly the stated target; the six §002.LEARN routes answer as declared. The 8 DEBT rows are recorded NOT_BUILT contract endpoints, printed by design
- [x] T056 Confirm `GET /api/areas` returns **42**, not the 819 the running container still holds in memory from start-up (FR-018, FR-023a, SC-001). — EVIDENCE (served): `GET http://127.0.0.1:8087/api/areas` -> **200**, `contract.total_areas = 42`, `len(areas) = 39` + `len(hel`

`d_back) = 3`; every id in `areas` carries an `href`, and `GET /api/areas/{first}` -> 200

- [x] T057 Exercise all six learning routes over HTTP against authored content. Everything so far is measured through the server's own loader **in-process**; the HTTP surface is unmeasured (plan.md gap 1; FR-006, FR-007, FR-009, FR-010, FR-012a, FR-013, SC-006, SC-009). — EVIDENCE (served): all six exercised over HTTP with `X-Session: audit-004-probe` on 2026-09-08: `GET /api/areas/{a}/lessons` 200 (7 lessons), `GET /api/areas/{a}/lessons/{l}` 200 (carries `prev/next/position`), `POST /api/areas/{a}/lessons/{l}/state` 200 (`complete_of_required` 0 -> 1), `GET /api/areas/{a}/materials` 200 (+ `?kind=<unknown>` -> **400**, T022 on the wire), `GET /api/areas/{a}/assessment` 200 (`available:false` with the missing-lesson list), `POST /api/areas/{a}/assessment/submit` -> **403** with the availability envelope, never a zero score
- [x] T058 [P] Register the new gates in `workshop/platform/gates/check-registry-002.tsv` and the pipeline gates in their registry (FR-026, SC-013). **Evidence needed**: `verify-check-registry-002.sh` currently rc 0 at `checks=94 debt=3`. — EVIDENCE (source): `bash workshop/platform/gates/verify-check-registry-002.sh` rc **0** — `checks=103 debt=3 missing=0 unreadable=0 proof_fail=0` (was 94/3); the new rows are `check-registry-002.tsv:444-453` (`G-AREA-NAME`, `G-LEARN-PUBCONSIST`, `G-LEARN-GATE`, `G-SERVED-PALETTE`, `G-SERVED-CONTRAST`, each with its `-proof` sibling) plus `G-AREA-TOPICALITY-classifier` and the `FR-004-obtainability` pair
- [ ] T059 [P] Add `workshop/curriculum/area-timelinks.json` and `workshop/curriculum/unpublished-areas.jsonl` to `.gitignore`, matching their siblings (FR-?).
- [ ] T060 [P] Re-run the full frontend suite and account for every delta against the 162/0/8 baseline (FR-?).
- [ ] T061 [REVIEW] **Human checkpoint** — read the 3 area documents failing publication review on 62, 54 and 58 uncited claim blocks. They stay failing until read; stamping them is the bypass (FR-002).
- [ ] T062 [REVIEW] **Human checkpoint** — confirm or replace the 70% `passPercent`, the mcq-2/short-3 point weights, the 91–95 prefix shift and the anchor attachment section. These four are authored schema choices, stated rather than measured (FR-008b, SC-017).
- [x]

T063 [SUBAGENT] Re-mine chapter 02 into the taxonomy, or record the coverage gap explicitly. A topic taught only there, in words the chapter-01 lexicon lacks, is invisible to every coverage figure in this feature (FR-005). — EVIDENCE: `workshop/docs/training/CURRICULUM-AREAS.md:108` records the gap explicitly and by name — chapter 02 and 02.01 contribute tier-A term matches only, no span-tier evidence and no chapter-02 coverage report has ever been built, so a topic taught only there is invisible to every figure on that page. This is the task's second arm; the re-mine itself was NOT done

- [ ] T064 Regenerate the 12 pre-existing documents' ingested `.sections.json`, or record that their corpus copy is stale. **Regenerating mints new pids and changes the searchable corpus** — a corpus decision, not a cleanup (FR-?).

- [ ] T066 [TDD] Assert **corpus stability during measurement** in every corpus-counting run (FR-030, FR-027) — fingerprint the enumerated set before and after the analysis passes, sharing the enumeration with the analysis by construction, and emit `undet` rows **naming the changed paths** when they differ. **Analysis finding F2: this requirement had zero task coverage.** It was promoted from a real incident — a gate reporting 12939 / 12939 / 13058 / 12968 for the same command on a tree another agent was editing. The umbrella's `scripts/verify-content-boundary.sh` already implements exactly this; reuse its approach rather than inventing one.

- [ ] T067 [TDD] [P] Assert **result stability** (FR-011, FR-026) in `workshop/platform/gates/prove-assessment-gate.sh` — submit, re-read, require a byte-identical result. **Analysis finding F4: FR-011 appeared only as a manual quickstart step, so nothing asserted it.**

- [ ] T068 [P] Detect **duplicate subjects** (FR-020, FR-026) in `workshop/pipeline/extract/verify_curriculum_areas.py`, or record FR-020 as deferred with its reason. **Analysis finding F5**: an existing mutation catches one *area* in two files; nothing catches two areas covering one *subject*.

- [x] T069 [P] Report **question-bank coverage** on every run as a fraction of published areas (SC-004a, FR-008), currently 5 of 42. A tracked figure, never a floor. — EVIDENCE (source): `workshop/pipeline/extract/build_learning_catalog.py:506-510` prints the fraction on every run; dry run 2026-09-08 rc **0**: `areas considered 42 / with`

```
lessons 42 / with an assessment 30 / WITHOUT assessment 12
(each a CK021 finding; no question generated to hide one).
```
NOTE: SC-004a and Amendment A1 both say **5 of 42**; the measured figure is **30 of 42** (`verify_question_banks.py` rc 0, 30 banks / 234 questions / 597 citations resolved) — the spec figure is superseded

- [ ] T070 [P] Append the governing FR/SC identifiers to each task line in this file (SC-013). **Analysis finding F8**: 33 of 39 FRs and 14 of 18 SCs are cited nowhere by id, so **SC-013 ("100% of requirements map to a check") is mechanically unverifiable** — it can be argued but not computed. With ids present it becomes a `grep`.
- [ ] T065 [SUBAGENT] Investigate the redaction-pipeline report: one segment is marked `redacted: true` while the same sentence survives in the merged transcript. Reported, not acted on (FR-?).

Dependencies

```
Phase 1 (kit + strict format)
  └─> Phase 2 (T007 single decision)         ← the checkpoint
        ├─> Phase 3 US1  ─┐
        ├─> Phase 4 US2  ─┤  independent of each other
        ├─> Phase 5 US3  ─┤
        ├─> Phase 6 US4  ─┤
        └─> Phase 7 US5  ─┘
Phase 8 US6  — independent of all stories
Phase 9      — requires every writing stream to have STOPPED
(T054)
```

T007 blocks every story. Nothing else does: US1–US5 touch disjoint file sets and were in fact built concurrently.

T054 blocks T055–T057 and must run exactly once, after all writers stop. A restart mid-write measures a tree that is still moving.

Parallel Execution

Four streams with disjoint file sets, as actually run:

Stream	Writes	Tasks
Content	`workshop/docs/training/`, `workshop/`	T005, T006, T008, T024–T026, T031,

Stream	Writes	Tasks
	`curriculum/`, `workshop/pipeline/`	T039, T040, T063, T064
Backend	`workshop/platform/backend/`, `workshop/platform/gates/`	T007, T010–T014, T017–T023, T032–T034, T041, T058
Frontend	`workshop/platform/frontend/src/app/`	T015, T016, T028, T029, T035–T038, T042, T043
Theming	`workshop/platform/frontend/src/styles/`	T045–T048

Cross-stream rule: any stream measuring another stream's output must rebuild immediately before measuring and state which build it measured.

Implementation Strategy

MVP = **Phase 1** + **Phase 2** + **Phase 3 (US1).** A catalogue that is honest about what it holds. T007 alone took listed-but-unopenable areas from 817 to 3.

Increment 2 = **Phase 4 (US2)** — lessons and a gated test. This is the mission as stated, and its backend is done; T028/T029 are what remain.

Increment 3 = **Phases 5–7.** Anchors, materials, presentation.

Phase 9 is not optional polish. Until T054–T057 run, this feature is verified on disk and in process, and **unverified as served** — which is the exact defect class (D6, D8) it was written to eliminate. Reporting it complete before then would repeat the mistake inside the fix for it.

Status

Counted from this file by script, not transcribed from a draft. An earlier revision of this table said 44 done; the file said 37. **The file was right**, and the discrepancy is recorded rather than quietly corrected, because a task list whose summary disagrees with its own checkboxes is the same defect class as a catalogue whose list disagrees with its detail route.

	tasks	done	remaining
Phase 1 Setup	4	4	0
Phase 2 Foundational	5	5	0
Phase 3 US1	7	7	0
Phase 4 US2	13	12	1
Phase 5 US3	9	8	1
Phase 6 US4	6	4	2
Phase 7 US5	5	4	1
Phase 8 US6	4	4	0
Phase 9 Polish	17	7	10
Total	**70**	**55**	**15**

Re-audited 2026-09-08 against evidence, not against task text. 18 tasks were found already delivered and are ticked above with the run that proves each. Two of them were delivered at a DIFFERENT path than the task names (T015 in `features/areas/curriculum-model.ts`, not `core/knowledge.ts`) and one was closed by its second arm rather than its first (T063 records the gap; it does not re-mine chapter 02) — both are stated on the line rather than smoothed over. **Phase 9's headline is withdrawn: the work IS now proved as served** — `verify-server-unity.sh` rc 0 at PASS=43 FAIL=0, all six learning routes exercised over HTTP, and the palette and contrast gates read the served bundle. The five tasks that remain PARTIAL are left UNTICKED with what is missing named in the audit report: T028 (no prev/next in the client), T038 (the deep-link miss path in `transcript.component.ts` `applyPending()` gives up SILENTLY — its own comment says so), T060 (suite re-run green at **298/298 SUCCESS, 0 failed, 0 skipped** against a 162/0/8 baseline, but the +136/-8 delta is not itemised), T066 (a corpus fingerprint exists in `classify_area_topicality.py` and is gate-checked, but it is a SINGLE fingerprint — there is no before/after pair and no `undet` row naming changed paths) and T059 (both files are TRACKED — `git ls-files` matches them — so a `.gitignore` line would change nothing until they are untracked, exactly the umbrella's own `_site` defect).

Re-derive rather than trusting the table:

```
grep -c '^- \[' specs/004-authored-ai-areas/tasks.md    # 70
grep -c '^- \[x\]' specs/004-authored-ai-areas/tasks.md  # 55
```

Five tasks (T066–T070) were added by `/speckit-analyze` remediation, closing the three requirements that had zero coverage (FR-011, FR-020, FR-030) plus the two traceability findings. None is done.

55 of 70 tasks carry a captured run (re-audited 2026-09-08). The earlier headline — *"not one line of it has been proved as served"* — is **WITHDRAWN**: 7 of the 17 Phase 9 tasks now carry a served measurement, including the restart, the unity gate at PASS=43 FAIL=0 and all six learning routes on the wire.

Superspec bridge — where the bite-sized steps live

`/speckit-superspec-tasks` ran the `superpowers:writing-plans` skill over the remainder. Its output is an implementation blueprint at

`workshop/docs/superpowers/plans/2026-09-07-served-verification-and-presentation-instruments.md` (380 lines)

which decomposes the **unheld** remainder into bite-sized TDD steps with exact files, real commands and expected exit codes.

Why only the unheld remainder. The scope check found that 6 of the 28 open tasks — T015, T016, T028, T029, T036 (frontend) and T045 (theming) — are being worked by live agents right now. Writing bite-sized steps for work in flight would have duplicated an agent mid-task and produced two authorities for the same file. Those tasks keep their entries here and take their direction from their agents.

Blueprint task	tasks.md	What it produces
A1	(new)	`lib/served_css.sh` — one definition of "what is served", refusing with **2** rather than guessing
A2	(new)	`sample-served-surfaces.mjs` — samples what is **painted**, both colour schemes, no verdict
A3	T046, T048	

Blueprint task	tasks.md	What it produces
		`verify-served-palette.sh` + prover, 3 data mutations, mutation 0 the control
A4	T047, T048	`verify-served-contrast.sh` + prover, incl. a mutation that passes light and fails dark
A5	T058	Registry rows; `checks=94 -> 98`, debt unchanged
B1	T054	The single restart, gated on a tree that has stopped moving
B2	T056	The list/detail pair asserted **over HTTP**
B3	T057	SC-009 measured **on the wire**, which is the only place it counts
B4	T055	`verify-server-unity.sh` 37/6 -> 43/0
B5	T044, FR-023a	Presentation measured against the restarted stack, with the build id stated

Sampling is split from verdict on purpose. It is what lets every mutation be a **data** edit to a fixture rather than a code edit to a gate — the §1.1 requirement that makes a prover trustworthy.

One gap the blueprint's self-review found and did not paper over: FR-023a asks the *served catalogue* to declare its corpus revision, while Track A carries a build id for the *bundle*. Different artefacts. That is **T049**, it belongs to the backend stream, and the blueprint says so rather than letting A2 look like it covers it.

Execution order:

```
A1 -> A2 -> A3 --+
            A4 --+-> A5        buildable NOW, writers still
running
[all writers stop] -> B1 -> B2 -> B3 -> B4 -> B5   (B5 needs
A3+A4)
```